

YAKAIAH SOLAM

PRINCIPAL ENGINEER

Result-oriented and creative professional, targeting challenging role in **Safety Critical System/Software Engineering** with an organization of high reputation in **IT industry**.



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CORE COMPETENCIES

System and Software Lifecycle

Project Management

People Management

Scaled Agile Framework (SAFe)

Product Certification

Embedded Software Development

Model Based Design

Team Building & Leadership

Research and Innovation



SOFT SKILLS

Leadership ■ ■ ■ ■ ■

Analytical ■ ■ ■ ■ ■

Communicator ■ ■ ■ ■ ■

Strategic ■ ■ ■ ■ ■

Motivator ■ ■ ■ ■ ■

Team-oriented ■ ■ ■ ■ ■

Problem Solving ■ ■ ■ ■ ■

Decision Making ■ ■ ■ ■ ■



PROFILE SUMMARY

- Strategic professional with **18 years of rich and extensive experience in Aerospace Embedded Systems, Software, Products and Solutions – FAR23, FAR25 airworthiness.**
- Leading **4 scrum teams** (25 members) as a **Release Train Engineer** (RTE) in terms of PI/Sprint planning, estimations, Sprint plan review (Velocity, Capacity vs Loading), sprint execution, tracking, mitigating risks, resolving dependencies, running Scrum of Scrums (SOS), Sprint Report Review (Planned vs Actual Velocity, Predictability, Efficiency).
- Guiding and mentoring 4 scrum teams (25 members) to improve velocity and efficiency.
- Expertise in embedded C/C++ programming and Real Time Operating Systems (RTOS).
- Extensive experience with design and development of embedded software for complex real-time and multi-tasking systems.
- Strong experience in embedded application software design, development, unit testing and integration of multiple modules of various safety critical aerospace and defense projects to compliance with **DO-178B/C**.
- Expertise in validating system and software requirements to compliance with **ARP 4754A** and **DO-178B, DO-178C, DO-331**.
- Delivered complex products to all customers on committed dates; won recognition from many customers and internal teams.
- Experienced in developing software for high-end commercial products, defining product strategy and roadmap, leading highly motivated technical teams, managing customer interactions and third-party vendors.
- Steered the complete application development cycle from creation of Statement of Work (SOW) to post-delivery support of the product/project.
- Directed the development of the **technology roadmap** for the division by identifying desired future products, technology, and intellectual property (IP) required to develop those products.
- Exposure in preparation for **SRR** (System Requirement Review), **PDR** (Preliminary Design Review) and **CDR** (Critical Design Review) reviews with customers, safety critical systems (**ARP 4754A**) and software audits (SOI #1, SOI #2, SOI #3, and SOI #4), rapid prototype development and technical decisions
- Expertise in development of **certification plans** (PSAC, SDP, SVP, SCMP and Standards) to compliance with **DO-178B, DO-178C & DO-331** for safety critical systems and software.
- Expertise in working with **certification authorities (FAA/EASA)** and review deliverable artifacts.
- Expertise in system and software development to compliance with airworthiness standards **FAR 23** and **FAR 25**.
- Expertise in setting up hardware integration testing environment (Bench), debugging, unit testing, integration testing and reproducing customer reported issues.
- Expertise in all phases of **Software Development Life Cycle** (SDLC).
- Exposure in **Product Development Life Cycle** (Analyze, Proposal, Negotiate, Development, Release & Support).
- Practicing **Scaled Agile Framework (SAFe)** and owning **Release Train Engineer** (RTE) role.



ACHIEVEMENTS

- Led 4 Scrum teams (25 members) to meet Sprint goals and Improved estimations by 15%.
- Implemented ART based metrics (Velocity, Efficiency, Estimations).
- Worked on Software Development for Airbus A220, Bombardier Global 7500 and Global 8000 with **Collins Proline Fusion Architecture**.

EDUCATION

- 2019:** M.Tech. | Embedded Systems | Birla Institute of Technology & Science | Pilani
- 2006:** B.Tech. | Computer Science and Engineering | MGIT | JNTU | Hyderabad
- 2003:** Diploma | Computer Science and Engineering | SES-SN Murthy Polytechnic | Khammam

TECHNICAL SKILLS

Technical Tools: DOORS, JAMA, JIRA, SVN, Bitbucket, PVCS, Serena Dimensions CM, Source Insight, Cygwin, Beyond Compare, Green Hills Probe (debugger), JTAG, Simulink, Simulink Design verifier, Embedded Coder, Simulink Code Inspector and Polyspace, GNU compiler, MULTI IDE, Bitbucket, LDRA, Rapita, Coverity, Microsoft Visual C++ compiler.

Languages worked/known: C, C++, Multithreading, TCP/IP Socket Programming, Message Queues, Mutex, Semaphores, Shared Memory, POSIX, Data Structures & Algorithms, STLs, Design Patterns, SOLID Design Principles, OOPs concepts, Python Basics

Operating System: Linux, Lynx (RTOS), Green Hills Integrity (RTOS), DEOS (RTOS) and Windows, Freescale MPC5553 Microcontroller

Standards: DO-178B, DO-178C, DO-330, DO-331, DO-254, DO-160, ARP-4754A, ARP-4761, ARINC-429, ARINC-653

Protocols: RS-232, SPI, ARINC-429, ARINC-664, TCP/IP

Certification Agencies: FAA, EASA
Airworthiness Standards: FAR 23, FAR 25

PERSONAL DETAILS

Date of Birth: 08-Feb-1982
Languages Known: English, Telugu, Hindi
Address: Villa #882, Symphony Park Homes, Beeramguda, Hyderabad

- Worked on System and Software Development for Dassault Programs (F7X, F6X and F10X) with **Honeywell EPIC Architecture**, ADAU programs (SJ30, Dornier SeaStar) with **Honeywell APEX Architecture** and Boeing 787 Flight Controls.
- Developed Safety Critical Software Certification plans for SJ30, Seastar, Dassault F7X and F6X and summary results.
- Successfully **demonstrated** Safety Critical software audits/reviews SOI #1, SOI #2, SOI #3, and SOI #4 for SJ30, Dornier Seastar, Dassault F7X and F6X with minimal observations.
- Successfully **prototyped** Model-Based Development (MBD) Proof of Concept (POC) for Advanced Data Acquisition Unit (ADAU) by using MathWorks Tools (MATLAB, Simulink, SLDV, Simulink Test, SLVV, Embedded Coder, Code Inspector and Polyspace to compliance with DO-178C & DO-331 and implemented in Dornier Seastar.
- Designed, Developed, and maintained** efficient, reliable, and reusable **C++** code in multiple projects – SJ30, Dornier Seastar, Dassault F6X, F7X, F9X, F10X

WORK EXPERIENCE

Role	Tenure	Organization
Principal Engineer	Sept'22 – Till date	Collins Aerospace - Hyderabad
Senior Advanced Embedded Engineer	Jul'18 – Aug'22	Honeywell - Hyderabad
Technical Lead	Oct'15 – Jun'18	Honeywell - Hyderabad
Senior Software Engineer	Jul'08 – Sep'15	Honeywell - Bangalore
Software Consultant	Jun'06 – Jun'08	ECIL - Hyderabad

Key Result Areas:

- Leading the 25-member team (4 scrums) to develop the features and helped team members in resolving technical issues.
- Managing schedules, timelines, scope, along with working with multiple stakeholders and partners.
- Leading entire project lifecycle, planning, requirement definition, design, implementation, integration, quality assurance, and upgrades.
- Monitoring progress with respect to cost, timely execution of project and steering delivery for technical deliverables of concurrent project.
- Providing technical solutions, resolving customer reported issues and resolving hardware & software integration issues.
- Working on new products, Architectures, and technologies.
- Strong ability and skills in design, coding, debugging, problem-solving, and integration.
- Finding ways to increase productivity and cost reduction through cycle time reduction.

CERTIFICATIONS

Completed various certifications:

- Green Belt Certified and Mentor in 6 Sigma
- Aerospace Certification COE Expert Program
- Scaled Agile Framework (SAFe)

PROJECTS

Project #1: Cockpit Displays for Bombardier M170, Global 8000 & Airbus A220

Environment: C, LynxOS178 RTOS, LynxOS178 Compiler, Microsoft Visual C++ Compiler, Trace32 Debugger, SVN, DOORS, JAMA, Microsoft Visual Studio 2015, Simulink, Embedded Coder, LDRA, Rapita Rapicover and Coverity.

Duration: 18 Months (September 2022 – Till date)

Description: The scope of this program is to integrate the new features such as **ROAAS** (Runway Overrun Awareness and Alerting System), **RNP AR 0.1** (Required Navigation Performance Authorization Required), OPR (Open Problem Reports) Fixes, Customer Reported issue fixes, Deactivated Code fixes for Bombardier programs.

Roles and Responsibilities:

- Software development on **Collins Proline Fusion Architecture** using C, Simulink and LynxOS178 RTOS environment.
- Acting as **Technical Lead** Engineer for the program and responsible for all new design aspects and quality of the delivery.
- Owing **Release Train Engineer** (RTE) role & handling 4 scrum teams (25 members) and responsible for PI/Sprint planning, execution, driving continuous development, managing risks, and escalating dependencies.
- Working in the Scaled Agile Framework (SAFe).
- Participating in the daily scrums, Program Increment (PI) planning, completing planned tasks for the sprint, escalating dependencies.
- Define software requirements, customer interaction, Change Requests, project scoping, estimation, risk management, stakeholder management and quality management.
- Lead the technical discussions and tie-up with stakeholders in the USA, and India (internal and external).
- Development, Task planning, tracking issues, reviews and mentoring team members.
- Development of software high-level requirements, low-level requirements, and code.
- Auto-Code generation using Embedded coder and development of manual code.
- Participating in system requirements reviews, Software development life cycle reviews.
- Participation in unit level, software integration testing, debugging, peer reviews and bug fixing.

Project #2: Cockpit Displays for Dassault's Falcon F6X, F7X/F8X, F9X, F10X

Environment: C++, DEOS RTOS, Microsoft Visual C++ Compiler, Dimensions, DOORS, Source Insight, LDRA, Source Code verifier, Simulink, multi-processes, and multi-threading.

Duration: 4 Years (July 2018 – August 2022)

Description: The scope of this program is to integrate the new features such as SVS (Synthetic Vision System), CDTI, 2D/3D AMM (Airport Moving Map), CORA(INAV), Hazard 2.0, RDR 7000, aircraft specific CAS & Synoptics and to develop a common framework for all Dassault programs.

Roles and Responsibilities:

- Software development on PRIMUS EPIC Architecture using C, C++ and DEOS RTOS environment.
- Acting as **Technical Lead** for the program and responsible for all new design aspects and quality of the delivery.
- Define system requirements, customer interaction, Change Requests, project scoping, estimation, risk management, stakeholder management and quality management.
- Lead the technical discussions and tie-up with stakeholders in the USA, Europe, and India (internal and external).
- Development of certification documents (PSAC, SDP, SVP, SCMP and Standards) to compliance with DO-178B/DO-178C.
- Functional owner for flexible window management which is the challenge to meet with new modalities.
- Prototyped solution and demonstrated to the leadership and customer which is much appreciated.
- Development, Task planning, assignments, tracking issues, reviews.
- Development of software high-level requirements, low-level requirements, and code.
- Auto-Code generation using RTW and development of manual code.
- Participating in system requirements reviews, Software development life cycle reviews.
- Participation in unit level, system integration testing, debugging, peer reviews and bug fixing.

Project #3: Advanced Data Acquisition Unit (ADAU) for SJ30, Dornier Seastar

Environment: C, Green Hills Integrity RTOS, MULTI IDE, JTAG, MPC5553 Microcontroller, Dimensions, Source Insight, Simulink, MATLAB, Simulink, System development, system testing

Duration: 4 Years (September 2014 – June 2018)

Description: The scope of this program is to develop brand new functionalities for aircraft specific CAS and Synoptics by acquiring different data from different aircraft sensors and providing processed data to PRIMUS APEX Architecture.

Roles and Responsibilities:

- Software development using C and Green Hills Integrity RTOS environment.
- Acting as Software Lead Engineer for the program and responsible for all new design aspects and quality of the delivery.
- Analyzing customer requirements, System changes, Problem Action Report (PARs) and Change Requests (CRs)
- Development of certification documents (PSAC, SDP, SVP, SCMP and Standards) to compliance with DO-178B/DO-178C.
- Providing trainings to the team on certification documents and its importance in the software development life cycle.
- Mentoring team members.
- Development of estimation model for ADAU programs.
- Involvement in rapid prototypes, technical decisions, Project Planning.
- Participation in Quotes estimations (QE) for new programs.
- Model development for customer requirements using Model-Based Development (MATLAB, Simulink).
- Development of simulation test cases for Models to ensure correctness of model with respect to the customer requirements.
- Development of software high-level requirements, low-level requirements, and code.
- Auto-Code generation using RTW and development of manual code.
- Participating in system requirements reviews, Software development life cycle reviews.
- Participation in unit level, system integration testing, debugging and bug fixing.
- Coordinating with Process team, Certification team and PDQE for the certification documents reviews and closure.

Project #4: Boeing 787 Flight Control System

Environment: C, Green Hills Integrity RTOS, MULTI IDE, Simulink, Source Insight, system Dev/Testing.

Duration: 6 Years (July 2008 – August 2014)

Description: The scope of this program is to develop brand new functionalities for Boeing 787 flight controls. Input processing (IP) and output processing (OP) are responsible to input acquisition, processing, and outputting.

Roles and Responsibilities:

- Analyzing customer problem reports (PR), internal problem reports (IPR) & system change request (SCR), identifying system level impact and investigation on internal problems which are found during flight testing.
- Interacting with lead system engineers (LSE) on PR/IPR/SCR and getting their concurrence
- Providing estimates for PR/IPR/SCR to the project engineer (PE) by using BOE (3Pt estimation model) and getting approval and prioritize tasks based on program schedules and commitments.
- Handling team of 3 members, work assignments, tracking, escalations, maintaining ABM
- Participation in System & Software Review Board (SSRB) meeting to approve system and software impacts.
- Writing system level requirements to comply with the customer requirements (PR/SCD) to meet ARP4754/ARP4754A.
- Writing software high-level requirements to comply with the system requirements (SRS/SCD) to meet DO-178B/C.
- Writing software low-level requirements and developing MATLAB/Simulink Models to comply with the software high level requirements to meet DO-178B/C.
- Writing C code and generating code from models using RTW as per the software design requirements, testing the code by creating private builds, configuration in CM21, creating review packet and closure.
- Testing system level functionalities by creating builds on target hardware Hardware/Software Integration testing.

- Providing co-ordination to the software/V&V teams.
- Participation in the regression analysis (Top-down, Bottom-up and Horizontal)
- Participating in system/software/design/code reviews and moderations
- Performing system checkouts.
- Performing Integration Testing for Signal processing (SP) partition.
- Preparation in Load Description Document (LDD) to deliver the load content to the customer.
- Involvement in the SOI #2 and SOI #3 (stages of involvement) audits.
- Involvement in the certification activities like updating LDD, VDD, validation matrix, verification matrix.

Project #5: Command Control System (CCS) for Brahmos Missile

Environment: C++, Linux, Lynx RTOS, GNU compiler, multi-threading, mutex, semaphores, message queues, shared memory, socket programming

Duration: 2 Years (June 2006 – June 2008)

Description: The scope of this program is to develop brand new functionalities for command control system for Brahmos missile for India defense. It involves different models like Commander Console (CC), Operator Console (OC), Communication Server (CS), Database Management system (DBS).

Roles and Responsibilities:

- Software development in C++ on Linux operating system with multi-process and multi-threaded application.
- Development of software high-level requirements, low-level requirements, and code.
- Integration of Software with the mainline build on Lynx RTOS
- Participation in unit level, integration testing, debugging and bug fixing.
- Designing of GUI using Sun Studio X-Designer tool.
- Mentoring new engineers.